Cell Tuition
1-1 PRIVATE TUTORS

ALGEBRA

Collecting Like Terms

Name _____ Date _____ No calculator Version 1

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KEY WORDS term *(tūnga)* · like terms *(tūnga ōrite)* · variable *(taurangi)* · simplify *(whakamāmā)*

A One variable ●○○

Remember: Like terms have exactly the same letters. Add or subtract the numbers in front: $3x + 5x = 8x$. Remember that x means $1x$.

1. Simplify.

a) $3x + 5x$ = _____ **b)** $7a - 2a$ = _____ **c)** $4y + y$ = _____ **d)** $9b - 8b$ = _____

e) $2m + 3m + 4m$ = _____ **f)** $10p - 4p$ = _____ **g)** $6x - 6x$ = _____ **h)** $5n - 2n + 4n$ = _____

i) $k + k$ = _____ **j)** $8c - 3c - 2c$ = _____ **k)** $12d - 7d$ = _____ **l)** $x + 9x$ = _____

B Two variables ●●○

Watch out: Only like terms can be collected. $3x + 2y$ cannot be made any simpler.

2. Simplify.

a) $3x + 2y + 4x$ = _____ **b)** $6m + 4n - 2m$ = _____ **c)** $4x + 3y + 2x + 5y$ = _____

d) $8a + 6b - 3a - 2b$ = _____ **e)** $10x + 7y - 10x$ = _____ **f)** $5p + q + p + 6q$ = _____

g) $9m + 5n - 4m - 2n$ = _____ **h)** $3a + 4b - 3b$ = _____ **i)** $2c + 2d + 2c$ = _____

C Terms with numbers ●●○

Watch out: A number on its own is not a like term with a letter term. $4x + 5$ is not $9x$.

3. Simplify.

a) $5a + 3 + 2a$ = _____ **b)** $7p + 5 - 3p + 2$ = _____ **c)** $2c + 9 + 5c - 4$ = _____

d) $5k + 1 + k + 6$ = _____ **e)** $6x + 8 - 2x - 3$ = _____ **f)** $4n + 10 - 4n$ = _____

g) $3y + 7 + 3y - 7$ = _____ **h)** $9m + 2 - 5m + 6$ = _____ **i)** $x + 4 + x + 4$ = _____

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D Products and quotients ●●●

> **Remember:** Multiply the numbers, then the letters: $3 \times 2a = 6a$ and $3a \times 2b = 6ab$. To divide, divide the numbers: $\frac{12a}{4} = 3a$.

4. Simplify.

a) $3 \times 2a$ = _____ **b)** $4x \times 5$ = _____ **c)** $a \times b$ = _____ **d)** $3a \times 2b$ = _____

e) $x \times x$ = _____ **f)** $5m \times 4n$ = _____ **g)** $6 \times 3y$ = _____ **h)** $2p \times 7q$ = _____

5. Simplify.

a) $\frac{12a}{4}$ = _____ **b)** $\frac{10x}{5}$ = _____ **c)** $\frac{8y}{2}$ = _____ **d)** $\frac{15m}{3}$ = _____

E Mixed practice ●●●

6. Simplify.

a) $6a + 3a - 2a$ = _____ **b)** $4x + 7 + 2x - 3$ = _____ **c)** $5m + 2n + 3m + 4n$ = _____

d) $9y - 9y + 4$ = _____ **e)** $2p + 3q + 6p - 3q$ = _____ **f)** $8k + 5 - 8k + 1$ = _____

g) $3x + x + x + 2$ = _____ **h)** $10a + 4b - 6a + b$ = _____ **i)** $7c + 2 - 4c + 9$ = _____

F Problems ●●●

7. Write an expression for each perimeter, then simplify it.

a) A triangle with sides $3x$, $4x$ and $5x$

b) A square with sides of $2n$

c) A rectangle with length $5a$ and width $2a$

★ Extension

8. Simplify $5x + 3y + 4 - 2x + 4y - 1$.

9. Tama says that $2a + 3b = 5ab$. Explain why he is wrong.

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Collecting Like Terms

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

KEY WORDS term *(tūnga)* · like terms *(tūnga ōrite)* · variable *(taurangi)* · simplify *(whakamāmā)*

A One variable ●○○

Remember: Like terms have exactly the same letters. Add or subtract the numbers in front: $3x + 5x = 8x$. Remember that x means $1x$.

1. Simplify.

a) $4x + 3x$ = _____ **b)** $9a - 5a$ = _____ **c)** $6y + y$ = _____ **d)** $7b - 6b$ = _____

e) $3m + 2m + 5m$ = _____ **f)** $12p - 7p$ = _____ **g)** $8x - 8x$ = _____ **h)** $6n - 3n + 2n$ = _____

i) $h + h$ = _____ **j)** $9c - 4c - 3c$ = _____ **k)** $11d - 6d$ = _____ **l)** $x + 8x$ = _____

B Two variables ●●○

⚠ **Watch out:** Only like terms can be collected. $3x + 2y$ cannot be made any simpler.

2. Simplify.

a) $2x + 3y + 5x$ = _____ **b)** $7m + 2n - 3m$ = _____ **c)** $3x + 4y + 5x + 2y$ = _____

d) $9a + 5b - 4a - 3b$ = _____ **e)** $12x + 5y - 12x$ = _____ **f)** $4p + q + p + 7q$ = _____

g) $8m + 6n - 5m - 3n$ = _____ **h)** $2a + 5b - 4b$ = _____ **i)** $3c + 3d + 3c$ = _____

C Terms with numbers ●●○

⚠ **Watch out:** A number on its own is not a like term with a letter term. $4x + 5$ is not $9x$.

3. Simplify.

a) $4a + 6 + 3a$ = _____ **b)** $8p + 4 - 2p + 3$ = _____ **c)** $3c + 8 + 4c - 5$ = _____

d) $4k + 1 + k + 7$ = _____ **e)** $7x + 9 - 3x - 4$ = _____ **f)** $5n + 12 - 5n$ = _____

g) $2y + 9 + 2y - 9$ = _____ **h)** $8m + 3 - 6m + 5$ = _____ **i)** $x + 5 + x + 5$ = _____

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D Products and quotients ●●●

> **Remember:** Multiply the numbers, then the letters: $3 \times 2a = 6a$ and $3a \times 2b = 6ab$. To divide, divide the numbers: $\frac{12a}{4} = 3a$.

4. Simplify.

a) $4 \times 3a$ = _____ **b)** $2x \times 6$ = _____ **c)** $x \times y$ = _____ **d)** $4a \times 2b$ = _____

e) $y \times y$ = _____ **f)** $3m \times 6n$ = _____ **g)** $5 \times 4y$ = _____ **h)** $3p \times 5q$ = _____

5. Simplify.

a) $\frac{15a}{3}$ = _____ **b)** $\frac{12x}{6}$ = _____ **c)** $\frac{18y}{2}$ = _____ **d)** $\frac{20m}{4}$ = _____

E Mixed practice ●●●

6. Simplify.

a) $7a + 2a - 3a$ = _____ **b)** $5x + 8 + 3x - 2$ = _____ **c)** $4m + 3n + 2m + 5n$ = _____

d) $6y - 6y + 7$ = _____ **e)** $3p + 2q + 5p - 2q$ = _____ **f)** $9k + 4 - 9k + 2$ = _____

g) $2x + x + x + 5$ = _____ **h)** $12a + 3b - 7a + b$ = _____ **i)** $8c + 3 - 5c + 6$ = _____

F Problems ●●●

7. Write an expression for each perimeter, then simplify it.

a) A triangle with sides $4x$, $5x$ and $6x$

b) A square with sides of $5n$

c) A rectangle with length $8a$ and width $5a$

★ Extension

8. Simplify $5x + 3y + 4 - 2x + 4y - 3$.

9. Tama says that $2a + 3b = 5ab$. Explain why he is wrong.

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